Skyler AI vs. Existing Conference Papers — Gaps and Outcomes

The papers

	Abdelhady & Shaker (2025, IEEE ACDSA)	**La Min Ei & Yongsiriwit (2025, IEEE InCIT)**	**Al-Subaiey et al. (2024, Computers & Electrical Engineering)**
Model	BERT + RoBERTa + DistilBERT ensemble	Llama 3.2, few-shot	Unspecified classifier + XAI layer
Task	Spam classification	Phishing classification	Phishing classification, with explanations
Reported metrics	**98.46% accuracy, 98.47% F1**, 32.82 sec to train	Classification performance on a labeled set (paywalled results)	**0.99 F1**, claimed "real-time web-based" deployment
What they measure	Classification quality, offline, once	Same	Classification quality + explanation method (XAI "to enhance user trust"), for one label
What they don't measure	Request-time serving behavior; whether a multi-signal decision is correct; privacy guarantees	Same	Same — "real-time" is claimed, zero latency numbers given; explanations cover one label, not a combined rule; no privacy claim or verification

All three make a production/real-time/trust claim (training speed, lightweight model choice, or "real-time web-based") without measuring the thing that claim depends on. None measures or proves a privacy guarantee.

The gap

Across all three: a model is validated for accuracy, and a production-readiness claim is made on top of it — but none of the three measures serving-time cost, none measures whether a *combined, multi-signal* decision (not just one label) is correct, and none measures or proves anything about where user data goes during classification, despite privacy being a stated reason to prefer small local models over cloud LLMs in this space.

Skyler AI's outcome, measured against that same gap

Axis	What the papers report	Skyler AI, measured
Classification quality	98.46% / 98.47% F1 (spam, ensemble); 0.99 F1 (phishing, unspecified model)	**89.91% accuracy / 90.27% F1** on the real, official SST-2 benchmark — different task (sentiment, not spam/phishing), stated plainly
Request-time serving cost	Not measured by any of the three	**252–310 ms per request** of model-reload overhead identified and eliminated; live `classifyMs` telemetry on the real endpoint
Per-email classification cost	Not measured by any of the three	**~22–26 ms/email** at realistic 200-email scale, no LLM call per email
End-to-end decision accuracy (sentiment AND keyword/semantic match combined)	Not measured by any of the three	**94.4% accuracy, 100% precision, 83.3% recall, 90.9% F1** — measured on a labeled 90-email test set against the real, deployed system
Explainability	Al-Subaiey et al. explain one label with an unspecified XAI method; the other two give none	**Every kept/discarded email now returns why**: sentiment label, matched keyword (if any), semantic-match flag — for the actual combined decision, live on the endpoint (`filterEmailsWithReasons`)
Privacy / data locality	Not measured or proven by any of the three, despite being a stated motivation for small local models over cloud LLMs	**Proven, not claimed**: a network-call interceptor patched into Node's http/https/net/tls/fetch before model load records zero network calls across model loading, sentiment analysis, and semantic embedding — verifiable by anyone who runs `verifyNoNetworkEgress.ts`
Latency predictability under real load	Not measured by any of the three	**1.2x–8.2x** reduction in latency variance; worst-case spikes bounded from 11–14 sec down to ~8 sec

Bottom line

On the one axis that's directly comparable — single-model classification accuracy — Skyler AI does not beat the best IEEE number (89.91% vs. 98.46%/0.99 F1), and it is a different task in each case. That is stated plainly. On every other axis — request-time serving cost, per-email cost, combined-decision accuracy, explainability of that combined decision, and privacy — **none of the three papers reports a number or a proof at all**. Skyler AI does, against a real, deployed system, with the method open for anyone to re-run.